CHAPTER 60

# Tuberculosis

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*Mycobacterium tuberculosis* (MTB) is the causative bacteria for the development of Tuberculosis (TB) and it is part of a complex of organisms including *M. boivs* (reservoir cattle) and M. africanum (Reservoir human).

## Epidemiology

There were an estimated 9.2 million new cases reported worldwide and it is estimated that 14.4 million prevalent cases, around one-third of the world's population has latent TB and 1.5 million deaths attributable to TB. The majority of cases occur in the world's poorest nations, who struggle to cover the costs associated with management and control programmes. The resurgence of TB has been largely driven by HIV disease and other immunocompromised diseases.

## Etiogy

1. M. Bovis infection arises from drinking non-sterilised milk from infected cows.
2. M. tuberculosis is spread by the inhalation of aerosolised droplet nuclei from other infected patients.

## Pathology and pathogenesis

Inhalation of aerosolised droplet nuclei from other infected patients

↓

The organisms lodge in the alveoli

↓

Initiate the recruitment of macrophages and lymphocytes

↓

Macrophages undergo transformation into epithelioid and Langhans cells

↓

Which aggregate with the lymphocytes to form the classical tuberculous granuloma

↓

Numerous granulomas aggregate to form a primary lesion or 'Ghon focus' (a pale yellow, caseous nodule, usually a few mm to 1-2 cm in diameter)

↓

It is situated in the periphery of the lung

↓

Tuberculosis (TB)

## Mode of Tuberculosis (TB) spread

Spread to the hilar lymph nodes

↓

Develops similar pathological reaction

↓

The combination of a primary lesion and regional lymph nodes

↓

Primary complex of Ranke

↓

Reparative processes encase the primary complex in a fibrous capsule

↓

This lesion eventually calcifies (limit the spread of bacilli-so-called latent T.B)

↓

It is clearly seen on a chest X-ray

## Latent T.B

Lymphatic or haematogenous spread may occur before immunity is established

↓

Seeding secondary foci in other organs

↓

(Lymph nodes, serous membranes, meninges, bones, liver, kidneys and lungs)

↓

The estimated lifetime risk of developing disease after primary infections is 10%

↓

Which may lie dormant for years

## Clinical features: Pulmonary disease

1. Chronic cough
2. Haemoptysis
3. Fever
4. Night sweats
5. Malaise
6. Loss of appetite
7. Loss of weight
8. Unresolved pneumonia

9. Asymptomatic(diagnosis on chest x ray)
10. General debility
11. Exudative pleural effusion

### Military TB

1. Blood-borne dissemination gives rise to miliary TB, which may present acutely.
2. Characterised by 2-3 weeks of fever, night sweats, anorexia, weight loss and a dry cough.
3. Hepatosplenomegaly may develop and the presence of a headache may indicate coexistent tuberculous meningitis.
4. Auscultation of the chest is frequently normal, although with more advanced disease widespread crackles are evident.
5. Fundoscopy may show choroidal tubercles.
6. The classical appearances on chest X-ray are of fine 1-2 mm lesions (millet seed) distributed throughout the lung fields, although occasionally the appearances are coarser.
7. Anaemia and leucopenia reflect bone marrow involvement.

### Clinical features : Extra pulmonary disease

Extra pulmonary tuberculosis accounts for about 20% of cases in those who are HIV-negative but is more prevalent in HIV-positive individuals.

### Lymphadenitis

1. Lymph nodes are the most common extra pulmonary site of disease.
2. Cervical and mediastinal glands are affected most frequently, followed by axillary and inguinal; more than one region may be involved.
3. Disease may represent primary infection, spread from contiguous sites or reactivation.
4. Supraclavicular lymphadenopathy is often the result of spread from mediastinal disease.
5. The nodes are usually painless and initially mobile but become matted together with time.
6. When caseation and liquefaction occur, the swelling becomes fluctuant and may discharge through the skin with the formation of a 'collar-stud' abscess and sinus formation.
7. Approximately half of cases fail to show any constitutional features such as fever or night sweats.
8. The tuberculin test is usually strongly positive.
9. During or after treatment paradoxical enlargement, development of new nodes and suppuration may all occur but without evidence of continued infection, rarely, surgical excision is necessary.

### Gastrointestinal disease

1. Upper gastrointestinal tract involvement is rare
2. Ileocaecal disease accounts for approximately half of abdominal TB cases.
3. Fever, night sweats, anorexia and weight loss are usually prominent and a right iliac fossa mass may be palpable.
4. Up to 30% of cases present with an acute abdomen.
5. Ultrasound or CT may reveal thickened bowel wall, abdominal lymphadenopathy, mesenteric thickenings or ascites.

### Pericardial disease

1. Disease occurs in two forms: Pericardial effusion and constrictive pericarditis.
2. Fever and night sweats are rarely prominent
3. The presentation is usually insidious with breathlessness and abdominal swelling.
4. Pulsus paradoxius, a raised JVP, hepatomegaly, prominent ascites and peripheral oedema are common to both types.
5. Pericardial effusion is associated with increased pericardial dullness and a globular enlarged heart on chest X-ray
6. Constriction is associated with a raised JVP, an early third heart sound and, occasionally, atrial fibrillation; pericardial calcification occurs in around 25% of cases.

### Central nervous systems disease

1. Meningeal disease represents the most important form of central nervous system TB.
2. Unrecognized and untreated, it is rapidly fatal.

### Bone Tuberculosis

1. The spine is the most common sites for bony TB (Pott's disease)
2. Which usually presents with chronic back pain and typically involves the lower thoracic and lumbar spine.

3. The infection starts as a discitis and then spreads along the spinal ligaments to involve the adjacent anterior vertebral bodies, causing angulation of the vertebrae with subsequent kyphosis.
4. Paravertebral and psoas abscess formation is common and the disease may present with a large (cold) abscess in the inguinal region.

### Joint Tuberculosis

1. TB can affect any joint, but most frequently involves the hip or knee.
2. Presentation is usually insidious with pain and swelling
3. Fever and night sweats are uncommon.

### Genitourinary disease

### Renal tract Tuberculosis

1. Fever and night sweats are rare
2. Mildly symptomatic for many years Hematuria, frequency and dysuria are often present
3. Sterile pyuria found on urine microscopy and culture.
4. In women, infertility from endometritis, or pelvic pain and swelling from salpingitis or a tubo-ovarian abscess occur occasionally.
5. In men, genitourinary TB may present as epididymitis or prostatitis.

### Physical examination

1. Lymphadenopathy in TB occurs as painless swelling of 1 or more lymph nodes.
2. Lymphadenopathy is usually bilateral and typically involves the anterior and posterior cervical chain or supraclavicular nodes.
3. Patients with pulmonary TB have abnormal breath sounds, especially over the upper lobes or involved areas.
4. Crackles/ Crepitations/Rales or bronchial breath signs may be noted, indicating lung consolidation.

   Signs of extrapulmonary TB differ according to the tissues involved.

- Confusion
- Coma
- Neurologic deficit
- Chorioretinitis
- Lymphadenopathy
- Cutaneous lesions

### Diagnosis

1. The presence of an otherwise unexplained cough for more than 2-3 weeks, particularly in an area where TB is highly prevalent, or typical chest X-ray changes should prompt further investigation.
2. Direct microscopy of sputum (induced with nebulised hypertonic saline if not expectorating) at least 2 but preferably 3 (early morning samples). The probability of detecting acid-fast bacilli is proportional to the bacillary burden in the sputum (typically positive when 5000-10,000 organisms are present). Tuberculous bacilli are difficult to stain due to their substantial lipid-rich wall

   The most effective techniques are the Ziehl-Neelsen and rhodamine-auramine stains

   Rhodamine-auramine stains causes the tuberculosis bacilli to fluoresce against a dark background and is easier to use when numerous specimens need to be examined.
3. A positive smear is sufficient for the presumptive diagnosis of TB but definitive diagnosis requires culture.

   Smear-negative sputum should also be cultured, as only 10-100 viable organisms are required for sputum to be culture-positive.

### Blood investigations

Raised ESR, CRP, Anaemia etc

**Tuberculin skin test** (low sensitivity/ specificity; useful in primary or deep seated infection)

### Extrapulmonary

**Fluid examination** -CSF, ascitic, pleural, pericardial, synovial fluid

Tissue biopsy.

## MCQs

1. Which of these is the culture medium for Mycobacterium tuberculosis?
   a. Wilson blair medium
   b. Lowenstein–Jensen medium
   c. Mac Conkey's medium
   d. Dubo's culture medium
2. World Health Organization recommended a control strategy for TB known as:
   a. DOTS  b. Gene therapy
   c. Morphine  d. MCT
3. The first person who discovered Mycobacterium tuberculosis was
   a. Louis Pasteur
   b. Robert Koch
   c. Edward Jenner
   d. Hans Christian Gram

## Answer

| | | | | | |
|---|---|---|---|---|---|
| 1. | b | 2. | a | 3. | b |